

- ✓ Problem 1: Write a program that will give you in hand monthly salary after deduction on CTC - HRA(10%), DA(5%), PF(3%) and taxes deduction as below:

Salary(Lakhs) : Tax(%)

- Below 5 : 0%
- 5-10 : 10%
- 10-20 : 20%
- above 20 : 30%

```
# Write code here
salary = int(input("Enter the annual salary: "))
hra = 10/100
da = 5/100
pf = 3/100
fixed = salary * (hra + da + pf)
if salary < 500000:
    total = 0
elif salary >= 500000 and salary < 1000000:
    total = (salary-500000) * 10/100
elif salary >= 1000000 and salary < 2000000:
    total = 500000*(10/100) - (salary-1000000) * (20/100)
else:
    total = 500000*(10/100) + 1000000*(20/100) - (salary - 2000000)*(30/100)
final_salary = salary - total - fixed
final_salary = final_salary/12
print(f"Your monthly in hand salary is {final_salary}")
```

```
➦ Enter the annual salary: 1500000
Your monthly in hand salary is 106666.66666666667
```

- ✓ Problem 2: Write a program that take a user input of three angles and will find out whether it can form a triangle or not.

```
# Write code here
angle_1 = float(input("Enter the first angle of the triangle: "))
angle_2 = float(input("Enter the second angle of the triangle: "))
angle_3 = float(input("Enter the 3rd angle of the triangle: "))
sum = angle_1 + angle_2 + angle_3
if (angle_1 > 0 and angle_2 > 0 and angle_3 > 0) and sum == 180:
    print(f"These angle can make triangle")
else:
    print("These angles can't make triangle")
```

```
➦ Enter the first angle of the triangle: 90
Enter the second angle of the triangle: 90
Enter the 3rd angle of the triangle: 90
These angles can't make triangle
```

- ✓ Problem 3: Write a program that will take user input of cost price and selling price and determines whether its a loss or a profit.

```
# Write code here
cp = float(input("Enter the Cost price of the product: "))
sp = float(input("Enter the Selling Price of the product: "))
if sp > cp:
    print("It's a profit")
elif sp < cp:
    print("It's a loss")
else:
    print("No profit No loss")
```

```
➦ Enter the Cost price of the product: 34
Enter the Selling Price of the product: 34
No profit No loss
```

✓ Problem 4 : Write a menu-driven program -

1. cm to ft
2. km to miles
3. USD to INR
4. exit

```
# Write code here
print("This is a converter calculator")
print("Press 1 for centimetres to feet")
print("Press 2 for kilometres to miles")
print("Press 3 for USD to INR")
print("Press 4 to exit")
num = int(input("Press the number: "))
if num == 1:
    cm = int(input("Enter centimetres: "))
    result = (cm * 0.032)
    print(f" value in feets is {result}")
elif num == 2:
    km = float(input("Enter the value in kilometres: "))
    result = km * 0.621
    print(f" value in Miles is {result}")
elif num == 3:
    usd = float(input("Enter us dollar: "))
    result = usd * 85
    print(f" currency in INR is {result}")
elif num == 4:
    print("Exit the program")
else:
    print("Invalid number")
```

```
↩ This is a converter calculator
Press 1 for centimetres to feet
Press 2 for kilometres to miles
Press 3 for USD to INR
Press 4 to exit
Press the number: 5
Invalid number
```

✓ Problem 5 - Exercise 12: Display Fibonacci series up to 10 terms.

Note: The Fibonacci Sequence is a series of numbers. The next number is found by adding up the two numbers before it. The first two numbers are 0 and 1. For example, 0, 1, 1, 2, 3, 5, 8, 13, 21. The next number in this series above is 13+21 = 34

```
# Write code here
f_num = int(input("Enter the number till the number you want the febonacci series: "))
a = 0
b = 1
i = 0
while i <= f_num:
    c = a + b
    print(a)
    a = b
    b = c
    i = i + 1
```

```
↩ Enter the number till the number you want the febonacci series: 10
0
1
1
2
3
5
8
13
21
34
55
```

✓ Problem 6 - Find the factorial of a given number.

Write a program to use the loop to find the factorial of a given number.

The factorial (symbol: !) means to multiply all whole numbers from the chosen number down to 1.

For example: calculate the factorial of 5

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Output:

120

```
# Write code here
fact_num = int(input("Enter the number you want a factorial of: "))
i = 1
fact = 1
while i <= fact_num:
    fact = fact * i
    i = i + 1
print(fact)
```

```
↩ Enter the number you want a factorial of: 5
120
```

✓ Problem 7 - Reverse a given integer number.

Example:

Input:

76542

Output:

24567

```
num = 76542
num = num//10
print(num)
```

```
↩ 7654
```

```
# Write code here
num = 76542
rev_num = 0
while num > 0:
    l_num = num%10
    rev_num = rev_num * 10 + l_num
    num = num//10
print(rev_num)
```

```
↩ 24567
```

- Problem 8: Take a user input as integer N. Find out the sum from 1 to N. If any number is divisible by 5, then skip that number. And if the sum is greater than 300, don't need to calculate the sum further more. Print the final result. And don't use for loop to solve this problem.

Example 1:

Input:

30

Output:

```
# Write code here
n = int(input("Enter the number "))
i = 1
sum = 0
while i <= n:
    if i%5 == 0:
        i = i + 1
        continue
    else:
        sum = sum + i
        if sum > 300:
            break
    i = i + 1
print(sum)
```

```
➡ Enter the number 26
276
```

- ✓ Problem 9: Write a program that keeps on accepting a number from the user until the user enters Zero. Display the sum and average of all the numbers.

```
# Write code here
sum = 0
count = 0

while True:
    num = int(input("Enter a number (0 to stop): "))

    if num == 0:
        break

    sum += num
    count += 1

if count > 0:
    average = sum / count
else:
    average = 0

print(f"Sum of numbers: {sum}")
print(f"Average of numbers: {average}")
```

```
➡ Enter a number (0 to stop): 12
Enter a number (0 to stop): 23
Enter a number (0 to stop): 798
Enter a number (0 to stop): 56
Enter a number (0 to stop): 0
Sum of numbers: 889
Average of numbers: 222.25
```

- ✓ Problem 9: Write a program which will find all such numbers which are divisible by 7 but are not a multiple of 5, between 2000 and 3200 (both included). The numbers obtained should be printed in a comma-separated sequence on a single line.

Write code here

```
result = []
```

```
for number in range(2000, 3201): # 3201 because range upper bound is exclusive
    if number % 7 == 0 and number % 5 != 0:
        result.append(str(number)) # Convert number to string for joining later
```

```
print(", ".join(result))
```

➤ 2002,2009,2016,2023,2037,2044,2051,2058,2072,2079,2086,2093,2107,2114,2121,2128,2142,2149,2156,2163,2177,2184,2191,2198,2212,2219,2226,2

Problem 10: Write a program, which will find all such numbers between 1000 and 3000 (both included) such that

- ✓ each digit of the number is an even number. The numbers obtained should be printed in a space-separated sequence on a single line.

Write code here

```
result = []
```

```
for number in range(1000, 3001):
    num_str = str(number)
```

```
    if (int(num_str[0]) % 2 == 0 and
        int(num_str[1]) % 2 == 0 and
        int(num_str[2]) % 2 == 0 and
        int(num_str[3]) % 2 == 0):
```

```
        result.append(num_str)
```

```
print(" ".join(result))
```

➤ 2000 2002 2004 2006 2008 2020 2022 2024 2026 2028 2040 2042 2044 2046 2048 2060 2062 2064 2066 2068 2080 2082 2084 2086 2088 2200 2202 2

- ✓ Problem 12: Write a program to print whether a given number is a prime number or not

Write code here

```
num = int(input("Enter a number: "))
```

```
if num <= 1:
    print("Not a prime number")
```

```
else:
    is_prime = True
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break
```

```
if is_prime:
    print("Prime number")
else:
    print("Not a prime number")
```

➤ Enter a number: 5
Prime number

- ✓ Problem 13: Print all the Armstrong numbers in a given range.

Range will be provided by the user

Armstrong number is a number that is equal to the sum of cubes of its digits. For example 0, 1, 153, 370, 371 and 407 are the Armstrong numbers.

Write code here

```
a_num = int(input("Enter the number: "))
```

```
l = a_num
```

```
sum= 0
```

```
while a_num > 0:
```

```

num = a_num%10
sum = sum + num**3
a_num = a_num//10
if l == sum:
    print("Number is an Armstrong number")
else:
    print("Number is not an Armstrong number")

```

↩ Enter the number: 154
Number is not an Armstrong number

```

a_num = int(input("Enter the number: "))
l = a_num
sum = 0
n_digit = len(str(a_num))
while a_num > 0:
    num = a_num%10
    sum = sum + num**n_digit
    a_num = a_num//10
if l == sum:
    print("Number is an Armstrong number")
else:
    print("Number is not an Armstrong number")

```

↩ Enter the number: 9474
Number is an Armstrong number

✓ Problem 14 :Calculate the angle between the hour hand and minute hand.

Note: There can be two angles between hands; we need to print a minimum of two. Also, we need to print the floor of the final result angle. For example, if the final angle is 10.61, we need to print 10.

Input:

H = 9 , M = 0

Output:

90

Explanation:

The minimum angle between hour and minute hand when the time is 9 is 90 degrees.

```

# Write code here
H = int(input("Enter hour (0-12): "))
M = int(input("Enter minutes (0-59): "))

# Calculate positions
hour_angle = H * 30 + M * 0.5
minute_angle = M * 6

# Find the absolute difference
angle = abs(hour_angle - minute_angle)

# Minimum of the two possible angles
min_angle = min(angle, 360 - angle)

# Floor the result
print(int(min_angle))

```

↩ Enter hour (0-12): 9
Enter minutes (0-59): 0
90

